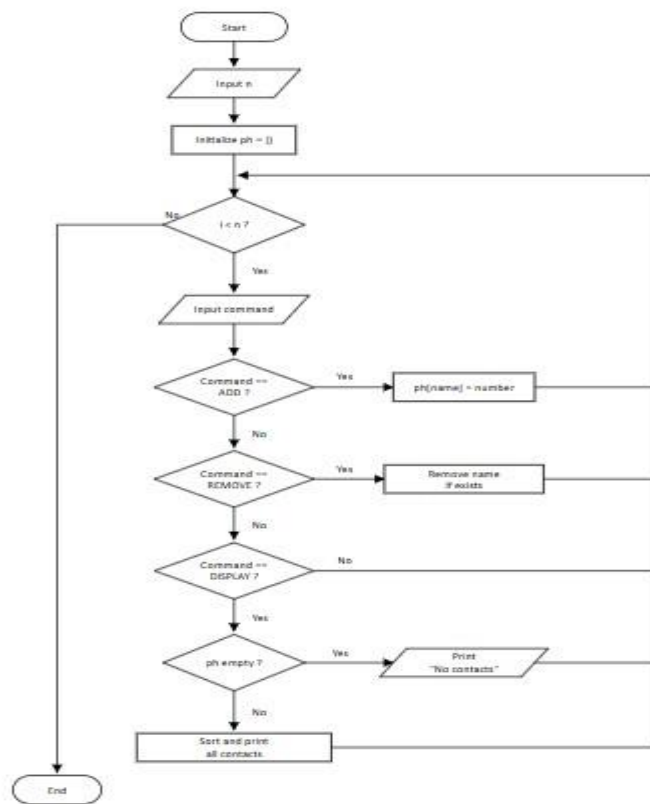


## EXPERIMENT 12.1.1 PHONE NUMBER DATABASE MANAGEMENT

### ALGORITHM:

- 1) Start
  - a) Input the number of operations n
  - b) Create an empty dictionary ph  
(This will store names as keys and phone numbers as values)
  - c) Repeat the following steps n times:
    - i) Read input command and split it into parts  
(Example: "ADD John 9876543210" → ["ADD", "John", "9876543210"]) ii)  
Check the operation type:
- 2) If command is "ADD":
  - a) Take name and phone number
  - b) Add to dictionary OR update if name already exists ph[name] =  
phone\_number
- 3) If command is "REMOVE":
  - a) Take name
  - b) Remove the contact if it exists  
(If not, do nothing)
- 4) If command is "DISPLAY":
  - a) If dictionary is empty:
    - i) Print "No contacts"
  - b) Else:
    - i) Sort all names in alphabetical order ii)  
For each name:  
(1) Print in format:  
name: phone\_number
- 5) End

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### 12.1.1. Phone Number Database Management

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

**Input Format:**

- First line contains an integer  $n$  representing the number of operations.
- Each of the next  $n$  lines contains a single operation in one of the following formats:
  - "ADD <name> <phone\_number>" where <name> is a string, <phone\_number> is a string of 10 digits.
  - "REMOVE <name>" - name is a string.
  - "DISPLAY" - no additional parameters.

**Output Format:**

- For display operation, print all contacts in alphabetical order by name in the format:

```
<name>: <phone_number>
```
- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

**Constraints:**

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

phoneDat...

Submit

```

1 # Phone Number Database Management
2
3 n = int(input())
4 contacts = {}
5
6 for _ in range(n):
7     command = input().split()
8
9     if command[0] == "ADD":
10         name = command[1]
11         phone = command[2]
12         contacts[name] = phone # add or update
13
14     elif command[0] == "REMOVE":
15         name = command[1]
16         if name in contacts:
17             del contacts[name]
18
19     elif command[0] == "DISPLAY":
20         if not contacts:
21             print("No contacts")
  
```

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YOUR PROGRAM HAS ENDED

